

SmartCare v0.4 - Domain Implementation Workbook

Week 7 student resource

1. UML-to-Code Trace

UML element	Python element	Implemented?	Notes
Patient class	class Patient:	Yes	Representing a patient
Patient ID	patient.id	Yes	Stores a unique ID for patient
Patient name	self.name	Yes	Stores the name of the patient
Patient registration	register()	Yes	Registers a patient
Practitioner class	class Practitioner	Yes	Representing a practitioner
Specialty	self.specialty	Yes	Stores practitioner's specialty.
Appointment class	class Appointment:	Yes	Represents an appointment.
Patient association	self.patient	Yes	Links an appointment to a patient
Book appointment	book()	Yes	Creates an appointment for a patient
Appointment time	self.time	Yes	Stores appointment time for a patient

2. Domain Invariants

Class	Invariant / rule	How protected
Patient	Patient ID has to be unique for all	ID validation required when creating a patient
Patient name	Patient cannot be left empty	It validates in patient class
Practitioner	Practitioner ID has to be unique for all	ID validation required when creating a practitioner
Practitioner name	Practitioner cannot be empty	It validates in practitioner class
Appointment	Appointment must have a valid patient.	Validate patient before booking.
Appointment	Appointment must have a valid date/time.	Validate date/time before booking.

3. Composition / Inheritance Decisions

Relationship	Decision	Rationale
Appointment – Clinic	Composition	Appointments are managed and related to the clinic
Doctor – Practitioner	Inheritance	A doctor can be considered as a practitioner
Clinic – Appointment	Composition	A clinic manages appointments, but an appointment is not a type of clinic.
Appointment – Patient	Composition	An appointment is for a specific patient
Appointment – Practitioner	Composition	An appointment is for a unique practitioner

4. AI Pair-Programming Record

AI contribution	Conforms?	Decision	Reason	Verification
Added patient to Appointment	Yes	Accept	Correct relationship	Checked object association
Added practitioner to Appointment	Yes	Accept	Correct relationship	Checked object association
Added status validation	Yes	Accept	Protects domain invariant	Tested invalid transition
Made status directly public	No	Reject	Breaks encapsulation	Reviewed access to status

5. Updated UML

Insert updated UML only if implementation revealed a justified design change. Explain every change.

- There was no change in the UML.